

VINJAYA BANDARA

vinjayabandara6@gmail.com | +60169446576 | linkedin.com/in/vinjayabandara | github.com/vinjaya04

EDUCATION

UNIVERSITY OF NOTTINGHAM

MSc (Hons) Mechanical, Materials and Manufacturing Engineering

Semenyih, Selangor

September 2024 – July 2028

Relevant modules: Thermodynamics and Fluid Mechanics, Materials and Manufacturing, Dynamics and Control

PROJECTS

Speedboat – CAE Modeling (Independent Project)

- Engineered 3D multi-body offshore speedboat on SolidWorks multi-plane lofts, guide curves and continuous curvature fillets to achieve smooth, hydrodynamic Class-A surface transitions across the outer hull and deck.
- Designed a cockpit and vessel subsystems, incorporating a 5-seat passenger layout, aft sunpad, custom helm console, gauge cluster, and integrated dual-triple exhaust ports.
- Defined realistic multi-body materials and generated high resolution CAD renders, applying photorealistic arine gelcoat, carbon fiber, teak decking and polished metals.

CAE Individual Design Project – Piston Modeling (SolidWorks)

- Designed and assembled a complete piston connecting rod system by modeling individual components including the piston, piston rings, gudgeon pin, little and big end bearings and connecting rod using SolidWorks.
- Ensured correct fit, alignment and assembly logic by applying appropriate mates, clearances and component relationships within the full assembly using SolidWorks assembly tools and engineering drawings

Mechanical Design Group Project – Pit Stop Challenge

- Designed a functional mechanical mechanism by developing and refining a concept to interact safely with an automated car using engineering design principles and SolidWorks.
- Controlled and safely halted the motion of a moving car by designing a pneumatically actuated barrier system and coordinating its timing with other subsystems using mechanical design principles and system integration planning.
- Translate the design into a physical prototype by manufacturing a pneumatic cylinder from the raw materials given, integrating it into the system using workshop machining processes and technical drawings.

Home Security system- Group Project (Arduino IDE)

- Built a smart-home prototype by designing and assembling a scaled house structure for system integration.
- Developed an embedded control system by programming Arduino-based sensor and actuator logic individually and integrating code modules using Arduino IDE.
- Implemented environmental monitoring functionality by coding and validating a temperature–humidity sensor module and integrating it into the final system using sensor libraries and serial output verification.

Centrifugal Clutch (Group Project)

- Designed a complete centrifugal clutch assembly by modelling individual components and defining their functional interactions using SolidWorks parametric CAD and mechanical design principles.
- Produced manufacturing-ready technical documentation by creating detailed part drawings and processed sheets for each component assigned using engineering drawing standards and structured process planning.

Variable Pitch Propeller (Individual Project)

- Designed and developed a propeller mechanism for a 25kW marine propulsion system incorporating a Scotch yoke actuation mechanism to achieve controlled blade pitch adjustment for forward and reverse operations.
- Performed engineering calculations for hydraulic piston sizing, pin shear strength and shaft loading to ensure safe and reliable operation under design requirements.
- Produced detailed 3D CAD models, engineering drawings and assembly layouts on SolidWorks including the propeller hub, hydraulic actuation system and support structures.

LEADERSHIP & ACTIVITIES

INTERNATIONAL STUDENTS' BUREAU (ISB)

Semenyih, Selangor

Head of Events

July 2026-present

- Leading and coordinating with members of the events team while working closely with the bureau heads to plan and execute well-organized events.
- Delegate responsibilities and support team members to ensure smooth and successful delivery.

STUDENT HELPER

Semenyih, Selangor

International Student Support

September 2026-October 2026

- Supported onboarding of 1000+ international students of more than 25 nationalities by guiding them through campus procedures, passport submissions and mandatory medical screening.

SRI LANKAN CULTURAL SOCIETY UNM

Semenyih, Selangor

Logistics Manager

April 2025 – April 2026

- Ensured smooth execution of society events by maintaining an up-to-date inventory list and tracking equipment availability using structured stock management and coordination practices.
- Delivered on-time event operations by coordinating transport and setup of logistics across multiple venues using effective scheduling and team coordination.
- Supported successful events and high attendance by sourcing, ordering, and managing timely delivery of food and event materials using vendor communication and deadline management.

SIPADAN HALL COMMITTEE UNM

Semenyih, Selangor

University of Nottingham

September 2023 - April 2024

- Enhanced resident engagement and hall participation by planning and coordinating residential events and community initiatives using clear communication and cross-team collaboration.
- Supported smooth hall operations by liaising with hall staff and residents to assist with logistical and operational needs using organizational and problem-solving skills.
- Improved residents well-being during peak academic periods by organizing and distributing exam-time care packages to residents using team coordination and time-sensitive planning.

SKILLS & INTERESTS

Technical: Python, MATLAB, MS Office apps, Arduino IDE, Cura

Interests: Thermodynamics, Dynamics, Aerospace, Formula 1, Fitness